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Identifying neuropathies through time series analysis of postural tests

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ABSTRACT

Background: In physical therapy, postural tests are frequently used to diagnose neuropathies, particularly in diabetic individuals. This study aims to develop a method based on the analysis of time series that allows discriminating between healthy and diabetic subjects with or without a neuropathic condition.

Research question: Do features obtained from time series corresponding to postural tests allow us to reliably discriminate between healthy, diabetic and neuropathic patients?

Methods: In this study, 32 people participated in the healthy, diabetic, and neuropathic categories (11, 9, and 12, respectively). The data was collected by positioning each participant on a Wii Balanced Board platform, under 8 different conditions. The analyzed time series are sensed by devices that capture variations in the subject's center of pressure when subjected to a test on different conditions over a short period of time. The method proposed considers statistical techniques used for characterizing the time series combined with machine learning techniques to classify the individual's profile into one of the three categories mentioned. The classification is supported by an underlying probabilistic model, based on the characteristics of the time series, generating average curves for each class, which are then used by the classification methods.

Results: The empirical results include classification models for each class, obtaining a performance (F-score) over 98%. In addition, other models considering the particular conditions to which the subject is exposed during the test are developed, revealing that the conditions of eyes open and eyes closed show the highest levels of discrimination to classify participants into one of the three class categories.

Significance: These results suggest a test protocol simplification and, at the same time, that the proposed method based on the analysis of the time series associated with the test used is highly predictive and may reliably complement or substitute a questionnaire-based diagnosis.

1. Introduction

Diabetes constitutes one of the main public health conditions concerns [1] and is one of the main causes of death worldwide [2]. Furthermore, diabetic neuropathy (DN) is one of the most prevalent complication associated with diabetes mellitus (DM), affecting up to 50% of individuals with diabetes [3]. Thus, DM and DN related impairment can increase the risk of fall, ulceration, and limb amputations, leading to decreased quality of life, disability, and institutionalization in aging population [2,4,5]. DN is characterized by damage in the neural tissue of peripheral nerves, retina, and vestibular system, compromising nerve conduction from the lower limbs, with a consequent detriment in spatial orientation, balance [6] and sensorimotor function

[7,8]. Although various assessing modalities are available for neuropathy diagnosis, there is no current consensus on which screening tests should be used in clinical practice.

In Health Sciences, body balance assessment is conducted through the instrumented analysis of static postural sway (i.e., posturography). The method consists in measuring the displacement of the center of pressure (COP) of an individual to obtain time series that represent static postural behavior over time. Previous research in postural control impairments related to DM in older adults, was associated with declines in reactive balance control [9]. DM has been related to subclinical imbalance disturbances before the manifestation of clinical signs of DN sensory impairment [10], particularly in challenging static balance conditions [11,12]. Moreover, abnormal postural parameters from force

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platforms were related to altered nerve velocity conduction in adults with DN [13,14].

Standard posturography methods obtain COP time series from force platforms, however, its high cost constitutes a constraint factor to their use in health services. As a low-cost alternative to force platforms, Nintendo Wii Balance Board (WBB) provide reliable data of body sway through the analysis of COP path length during static balance trials [4,6]. However, as commercial device designed for gaming, the WBB has some limitations related to overestimation and sample frequency inconsistency that may affect his performance as a measurement instrument [6,15,16]. In these context, methodological considerations must be taken in order to minimize the aforementioned constraints. Regardless of the instrument used, posture data is a reliable source of information, which allows identifying the presence of pathologies or differences between classes of subjects, through the changes caused to the static postural control. Given this context, instrumented methods based on posture assessments can be helpful in the early diagnosis of balance impairment in older people with diabetic neuropathy [12,17,18].

As an irreversible condition, DN early diagnosis is critical to prevent limb-treating conditions and morbidity. Although various assessing modalities are available for neuropathy diagnosis, there is no current consensus on which screening tests should be used in clinical practice. Then, standardized survey-type clinical evaluations are still required to label participants with their respective health condition. Given this context, instrumented posture assessments can be a useful tool for early diagnosis of balance impairment in older people with diabetic neuropathy in daily clinical care.

The use of machine learning techniques makes possible to expand knowledge and solve problems by studying data in different areas, such as in medicine, where it has allowed us to learn more about diseases affecting people [19]. Classic machine learning algorithms such as decision trees, support vector machines, artificial neural networks, short-term memory (LSTM), and others, have been applied to physiological data. The purpose is to improve the recognition of disorders such as arrhythmias [20], neuromuscular disorders [21], Parkinson's disease [22], and classification for diabetes mellitus disease conditions [23–25].

The problem addressed in this study is pathology detection and medical diagnosis through the study of individual postural control variables to determine how the body adapts when a health condition is present. This study consists of supporting healthcare professionals in the identification of diabetic neuropathy, using a statistical model supported with machine learning algorithms, for modeling the behavior of the posture data, and therefore support standardized survey-type clinical evaluations conducted in diabetic population.

As a result of this study, the following contributions can be identified:

1. The development of a statistical–computational method to classify participants, based on the profile of the signal generated during the postural test.
2. The formalization of an underlying probabilistic model from the data to feed different machine learning algorithms.
3. The development of models by the class and condition of the test used to make a comparative analysis in terms of the performance of each model, when classified by both category and a particular condition that is part of the postural test.
4. The generation of empirical evidence supporting the hypothesis that data from the time series allows classifying, with a high level of confidence, the condition of a subject undergoing a postural test.

The paper is structured as follows: In Section 2, materials and methods required to realize data preparation, modeling and experimentation are described. In Section 3, the results obtained from the classification models by classes and by tests are summarized. In Section 4, discussion of results are described. Finally, in Section 5 final conclusions are raised.

Table 1

Demographic participant data.

Variable	Healthy (n = 11)	Diabetes (n = 9)	Neuropathy (n = 12)
Age, years	60.27 ± 8.60	65.67 ± 11.10	66.13 ± 6.79
Weight, kg	73.77 ± 10.60	79.61 ± 7.84	68.44 ± 18.57
Height, m	1.62 ± 0.07	1.62 ± 0.08	1.61 ± 0.09
BMI, kg/m ²	28.03 ± 3.88	30.32 ± 2.59	26.24 ± 6.31
Diabetes diagnosis, years	–	21.89 ± 11.06	18.38 ± 13.94
Glucose, mg/dl	99.09 ± 13.44	149.78 ± 55.33	207.50 ± 82.01

2. Materials and Methods

2.1. Participants

Participants were thirty-two seniors (17 men and 15 women) and were classified into three groups: healthy group (HG; n = 11), diabetic group (DG; n = 9) and diabetic with diagnosed diabetic neuropathy group (DNG; n = 12). Inclusion criteria for DG were: Diabetes diagnosed over two years ago and independence in carrying out life activities according to Barthel Index Score. Participants were excluded if met any of the following exclusion criteria: Neurological conditions other than DN, vestibular disorders, uncontrolled hypertension, cognitive impairment according to the Minimental test, vision impairment, hypoglycemia, amputation, or any traumatic lower limb injuries in the past six months. Class labeling was performed by an evaluator using four standardized tests: the Barthel index [26], the Neurological Symptom Score (NSS) [27], the Diabetic Neuropathy Symptom (DNS) [28] and the tuning fork vibratory sensitivity test in 128 Hz and 256 Hz (in lower extremities) [29]. Participants demographics are shown in Table 1. All subjects gave informed consent to participate in the study. The experimental procedures were approved by the University of Antofagasta Ethics Committee, (CEIC-UA, Number 155/2018) and all requirements set out in the Declaration of Helsinki (1975, subsequently revised in 2008) were also met.

2.2. Experimental protocol

Four increasingly challenged balance conditions according to an instrumented version of the modified Clinical Test of Sensory Interaction for Balance were applied [30]. To challenge postural control, an unstable surface and dual task were used. The first consisted of a low-density foam surface that was positioned on the WBB. The second one consisted in a backward count from a random number indicated by an evaluator, in order to alter the participant's focus in the test. In total, eight different test conditions were performed by participants:

- Eyes open on stable surface (EO).
- Eyes closed on stable surface (EC).
- Eyes open on unstable surface (EOUS).
- Eyes closed on unstable surface (ECUS).
- Eyes open on stable surface with dual task (EODT).
- Eyes closed on stable surface with dual task (ECDT).
- Eyes open on unstable surface with dual task (EOUSDT).
- Eyes closed on unstable surface with dual task (ECUSDT).

Participants were instructed to stand barefoot and place both feet over reference marks in the WBB, to keep their hands alongside their body, and to stand as still as possible while looking straight ahead to a reference mark placed on the wall. Participants performed three 30-s trials on each condition with 20-s of sitting rest between them. To increase the number of COP time series, each participant was requested to complete each test three times. However, due to personal limitations of the participants to perform the evaluation, some of them performed less than three trials. Thus, 433 files were generated, 87 corresponding to HG, 145 to DG, and 201 to DNG.

2.3. Wii balance board data extraction

A WBB device contains four sensors in each corner: upper left (UL), upper right (UR), lower left (LL), and lower right (LR), which are used for calculating COP. Assuming that the distance between the sensors from left to right is 433 mm and that from the upper to lower sensors is 238 mm [31], then, COP estimation is shown in Eqs. (1) and (2):

$$COP_x = \frac{x}{2} \frac{(UR + LR) - (UL + LL)}{(UR + LR + UL + LL)} \quad (1)$$

$$COP_y = \frac{y}{2} \frac{(UL + UR) - (LL + LR)}{(UR + LR + UL + LL)} \quad (2)$$

where x and y correspond to the separation distances of the mediolateral and anteroposterior directions, respectively. The measure unit of the data is expressed in kilograms. During evaluation, data were transmitted telematically and stored in the lab computer, through BrainBloX software [32], which communicates both devices via Bluetooth. Finally, data were stored in the lab computer in a comma-separated values file (csv). The individual's class is not defined by the device, but rather by the evaluator, according to the aforementioned clinical tests.

Sampling data correction. The WBB mean sampling frequency is 63 Hz; however, it shows differences, according to three regimes: fast-paced (>100 Hz), moderate (≈ 54 Hz), and slow (≈ 35 Hz). To solve this problem, Audiffren & Contal [33] proposed a method known as Sliding Window Average with Relevance Interval Interpolation (SWARII), which consists of an interpolation window function that adjusts the values according to their highest concentration by weighting every point with its relevance interval. The mathematical model is defined as Eq. (3):

$$X_{SWARII}(s) = \frac{1}{\Delta} \sum_{t \in T_s(\Delta)} x_t I_{T_s(\Delta)}(t) \quad (3)$$

where I is the relevance interval, such that Δ is the window duration, s is the current instant of time and $T_s(\Delta)$ is a window centered around s . Then each observation x_t of the signal is fitted with the value of I . The value that I takes will depend on the separation of the current value and the rest.

2.4. Methodological steps

In this project, the work methodology considers adapting to the scientific study the six phases of the Cross-Industry Standard Process for Data Mining (CRISP-DM) guide [34], because CRISP-DM is still the most used in data science and data mining projects [35–37]. Fig. 1 represents the CRISP-DM methodology, grouped into analysis, design, evaluation and discussion of results. This process is iterative, incremental and considers all the tasks of exploration and preparation of data, model construction and analysis of final results.

2.5. Data preprocessing

After applying resampling method, all resulting time series were 25 Hz. Then, time series were truncated in 1.5 s at the start (38 samples). Subsequently, 20 s (500 samples) were selected, while the remaining samples were removed.

A normalization procedure was applied by subtracting the mean value of position to center all COP signals on the X (mediolateral) and Y (anteroposterior) axes around the value zero. Later, for the axes X and Y , a data matrix was created for each of the three groups: HG, DG and DNG. In these six resulting data matrices, each column represents a COP time series in some of the axes X , Y and each row represents an instant of time t_i , ranging from 0 to 499 indexes. Then, root mean squared (RMS) was calculated for each row and the result was added to a new data set, such that 500 values with their respective class were generated.

Table 2

Training–Testing data split.

Set	Healthy	Diabetic	Neuropathic	Total	%
Training	350	350	350	1050	70%
Testing	150	150	150	450	30%
Total	500	500	500	1500	100%

This process was repeated on velocity, acceleration and jerk, derived from the original position and on both axes. Empirically, and confirming with Mancini's work [38], from the second and third derivative (acceleration and jerk respectively), greater amplitude in time series were obtained. Therefore, velocity was not included in the final dataset. The above procedure generates a seven-column data frame (COP X, COP Y, Acceleration X, Acceleration Y, Jerk X, Jerk Y, and Class) for each class. Finally, the three new data frames created are merged into a new one generating a final data frame of 1500 rows (500 for each class). The representation of the final data frame processing is shown in Fig. 2.

2.6. Dataset split

For the modeling phase, two classification models were developed: first, data were split into 70% for training and 30% for testing in a stratified sampling, as shown in Table 2 (Hold-out). Second, a k-fold Cross-Validation operator was used, where data were split into 70% for training and validation purposes, and the remaining 30% for testing. No significant differences were obtained with values of k between 5 and 20 folds. Therefore, the one that obtained the highest performance is chosen, in this case $k = 10$ folds.

2.7. Root Mean Square (RMS)

As mentioned in Mancini's works [38,39], RMS is used because it is sensitive to change depending on the subject's conditions, i.e., age and health, related to falls or other conditions affecting him/her. It can be used on the position time series and their derivatives. More precisely, if z_{ij} represents the feature z of the j th individual observed at the instant t_i , the $RMS(z, t_i)$ at instant t_i is given by Eq. (4). This new time series is labeled with its corresponding class feature value.

$$RMS(z, t_i) = \sqrt{\frac{1}{N_{col}} \sum_{j=1}^{N_{col}} z_{ij}^2}, \quad (4)$$

where N_{col} is nothing but the number of columns of the corresponding matrix, whose construction was described previously in Section 2.5. Once the $RMS(z, t_i)$ is computed for each instant t_i , a new time series of length 500 is obtained. The process is repeated for each derived feature from COP. Fig. 3 is a graphical description of this procedure. At the right hand, we see the set of time series, of the same postural axis, concatenated and aligned at the same instant of time. The $RMS(z, t_i)$ is computed at each instant t_i and the value is added to a new data frame, with its respective class value.

2.8. Deriving the position

Derivatives calculation allows an analysis of behaviors that differ between the groups of subjects. To do this, the calculation of the second and third derivatives of the position is performed (acceleration and jerk, respectively). Acceleration is defined as the change in velocity of a body as Eq. (5) and jerk as the variation in its acceleration as Eq. (6), both in the time domain. Since they are vector quantities, they are obtained from the derivative of the position $\vec{r}$, represented as a COP discrete time series.

$$\vec{a} = \frac{d^2 \vec{r}}{dt^2} \quad (5)$$

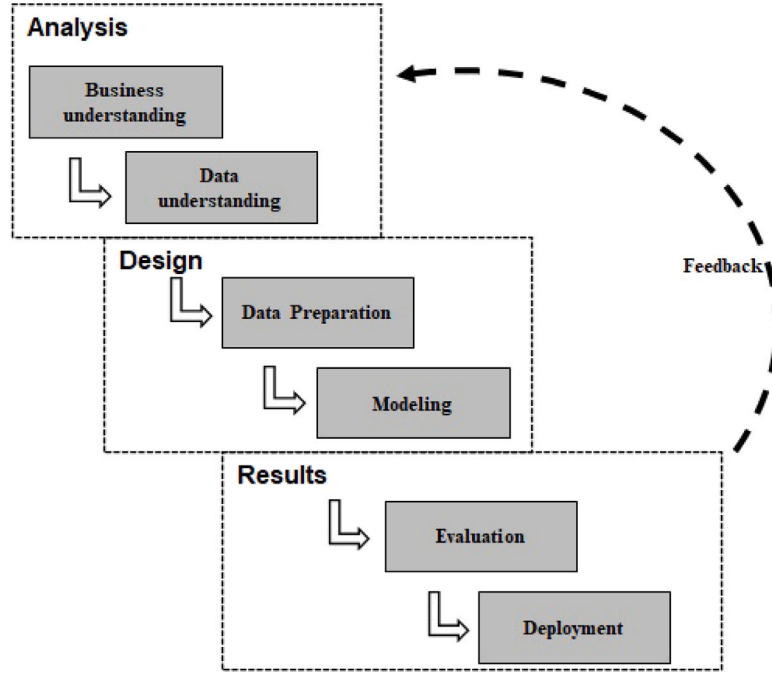


Fig. 1. An overview of the work methodology applied, based on the CRISP-DM guide [34].

	COP_x	COP_y	Acc_x	Acc_y	Jerk_x	Jerk_y	Class
0	0.681649	0.357277	0.433484	0.324126	0.457435	0.468179	Healthy
1	0.675154	0.351711	0.406740	0.281872	0.312508	0.282229	Healthy
2	0.670222	0.346279	0.422835	0.208643	0.255452	0.236001	Healthy
3	0.797282	0.703635	0.777565	0.417149	0.580636	0.389634	Diabetic
4	0.776744	0.695113	0.702759	0.459354	0.576341	0.467420	Diabetic
5	0.756577	0.688843	0.786323	0.634931	0.608914	0.384017	Diabetic
6	1.038499	0.757029	0.965177	0.684098	0.673923	0.460713	Neuropathic
7	1.045285	0.760989	1.056085	0.614598	0.619503	0.449208	Neuropathic
8	1.052244	0.766188	1.147122	0.521719	0.617937	0.437735	Neuropathic

Fig. 2. Slice of the first nine rows of the final data frame used for machine learning classifiers.

$$\vec{j} = \frac{d^3 \vec{r}}{dt^3} \quad (6)$$

Plotting the 433 time series at same time shows that position time series (i.e., COP) does not have a clear pattern to differentiate between groups. However, Fig. 4 shows that when deriving the position, a slight separation is perceived in the amplitude of each signal, mainly in the Y axis. When calculating acceleration and jerk, a separation between the groups can be perceived, showing difference in amplitudes in DNG, followed by DG and finally HG. Therefore, it is feasible to find differences between DNG and DG in acceleration and jerk, mainly in the Y axis. In the case of HG, the time series are less erratic in any of the features and they always maintain a dispersion lower than diabetic subjects.

2.9. A statistical model to represent classes by average curves

The following model is defined based on the behavior of the current data. Once COP series per participant and type of examination were

processed and derived on x and y axes, two new time series on each axis were considered: acceleration and jerk.

The exploratory analysis suggests that, at a subject level, the acceleration time series are stationary (in fact, the removal of the boundary conditions and discrete differentiation eliminate the trend patterns). Based on this evidence, it is proposed that each class has a particular equilibrium distribution, but different from each other. An important point to note is that the acceleration series appear centered and, therefore, the mean does not show a discriminatory behavior. However, the dispersion shows clear differences between the three classes. More formally, for each exam type, we denote by $(A_{x,\ell}^{(i)}(t_k), A_{y,\ell}^{(i)}(t_k))$, $k = 0, \dots, N - 1$ the vectors that correspond to the time series of the acceleration data on the x e y axes of the subject i of the class $\ell = 0, 1, 2$, a model of the type is proposed for each class, Eq. (7) for x axis and Eq. (8) for y axis:

$$A_{x,\ell}^{(i)}(t_k) = \sigma_{x,\ell}(A_{x,\ell}^{(i)}(t_{k-1}))\epsilon_{x,t_k} \quad (7)$$

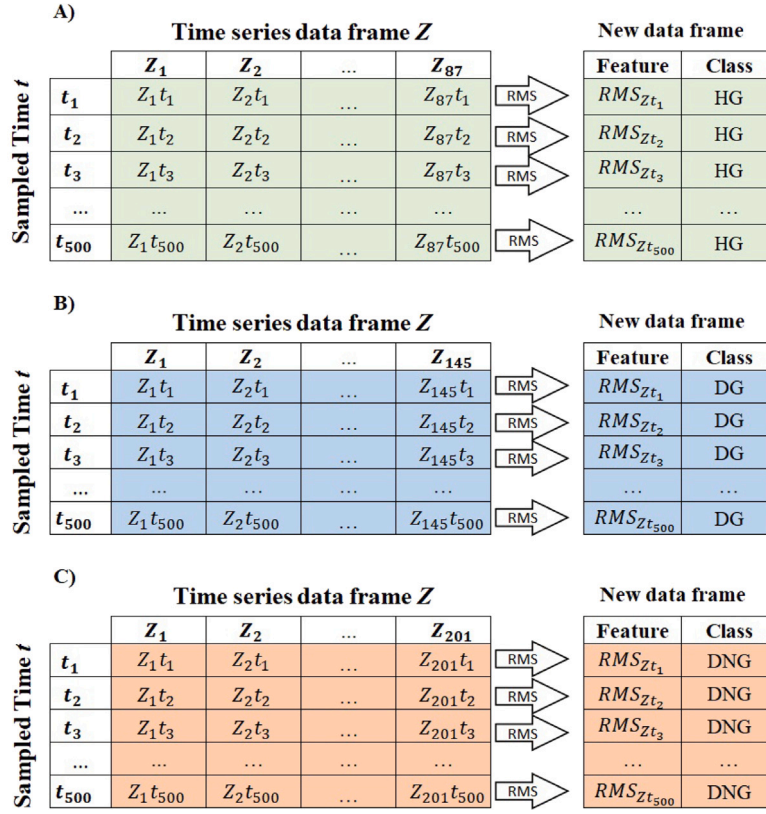


Fig. 3. Illustration of the RMS applied on a concatenated dataframe Z , which generates a new dataframe. It is applied to the 87 healthy time series (A), 145 diabetic time series (B) and 201 neuropathic time series (C).

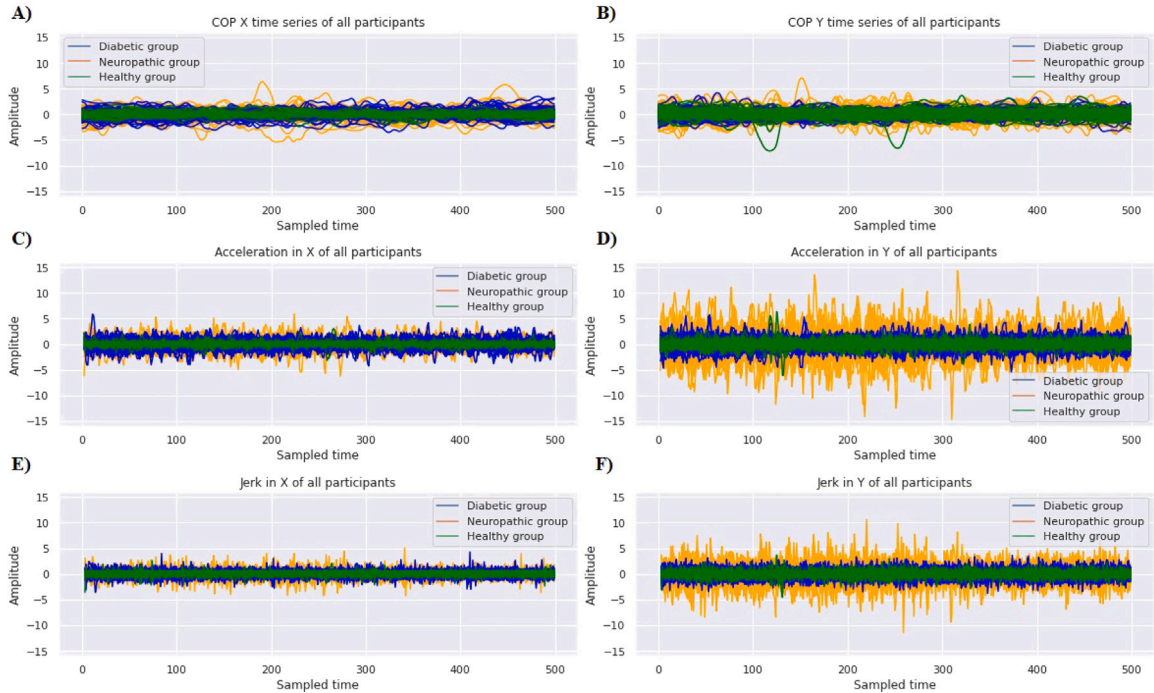


Fig. 4. All-time series set plotted, grouped by COP X (A), COP Y (B), Acceleration in X (C), Acceleration in Y (D), Jerk in X (E) and Jerk in Y (F) of all participants.

$$A_{y,\ell}^{(i)}(t_k) = \sigma_{y,\ell}(A_{y,\ell}^{(i)}(t_{k-1}))\epsilon_{y,t_k} \quad (8)$$

where $\sigma_{x,\ell}(\cdot)$, $\sigma_{y,\ell}(\cdot)$ are non-negative functions assumed different for each class, and ϵ_{x,t_k} , ϵ_{y,t_k} are two independent standardized white noise processes. This means $E[\epsilon_{x,t_k}] = 0$, $E[\epsilon_{x,t_k}^2] = 1$ (the same follows for

ϵ_{y,t_k}). Now, by taking the expected value for square of the acceleration on the axes x and y we have something in the form as Eqs. (9) and (10):

$$E[(A_x^{(i)}(t_k))^2] = E[\sigma_{x,\ell}^2(A_x(t_{k-1}))], \quad (9)$$

$$E[(A_y^{(i)}(t_k))^2] = E[\sigma_{y,\ell}^2(A_y(t_{k-1}))], \quad \ell = 0, 1, 2. \quad (10)$$

Now, if the time series is stationary, then $E[\sigma_{\ell}^2(A_x(t_{k-1}))]$ is constant for each class (the same applies for $A_y(\cdot)$). Since there are several participants in each class, this expected value can be estimated by taking the empirical averages at each instant according to Eq. (11):

$$\hat{C}_{\ell}^2 = \frac{1}{N_{\ell}} \sum_{k=1}^{N_{\ell}} (A_{x,\ell}^{(i)}(t_k))^2. \quad (11)$$

For each class we obtain an average curve corresponding to vector $(\hat{C}_{\ell}^2(t_k))_{k=0}^N$. These average curves are used as the characteristics of each class and are subsequently used as part of the classification algorithms.

2.10. Machine learning algorithms

Four machine learning algorithms were selected for classification [40]: Naive Bayes (NB), Artificial Neural Networks (ANN), K-Nearest neighbor (KNN) and Decision Tree (DTree). The NB classifier is based on Bayes' theorem, in which, given a set of training tuples labeled with their respective class, the conditional probability is calculated for each class value. Each tuple consists of a set of features $X = (x_1, x_2, x_3, \dots, x_n)$. If each of these belongs to one of the three classes of participants, then the classifier predicts the group it belongs by evaluating the maximum a posteriori hypothesis. The Naive Bayes model considers that there is a conditional independence among features given the class, which simplifies the model and reduces the computational cost when processing a large data set.

The ANN model consists of a set of interconnected nodes separated into an input layer, hidden layers, and an output layer. The features are separately entered in parallel, and as they go through them, the values are fitted by adding synaptic weights. These are adjusted iteratively until the least mean square error is reached or a number of predefined epochs are performed. A sigmoid activation function allows the adjustment of non-linear regressions and achieves the activation or inhibition of subsequent neurons. The model is trained for 200 epochs and a hidden layer with 6 neurons is used. The output layer consists of three nodes, one for each class value. A certain similarity to the conditional probability can be considered in that the output neuron with the least error among the three indicates the class of the input belongs to.

The K-NN model is used mainly in the area of pattern recognition and is based on similar learning tuples. The model allows searching for the pattern space of the k unknown values that are closest to the training values. The degree of "closeness" between two tuples $X_1 = (X_{11}, \dots, X_{1n})$ and $X_2 = (X_{21}, \dots, X_{2n})$ is based on a distance metric between them (e.g., Euclidean distance). When a new value is evaluated, the majority class of the closest k -neighbors is assigned. For this model, different values of k were tested empirically (from 3 to 6), being $k = 5$ the one that obtained the highest performance. However, no significant differences were obtained with other k values.

3. Results

3.1. Data exploration

When exploring the time series and their derivatives (Fig. 5), a noticeable separation of the three groups is visualized, when applying the underlying model previously described. This shows that HG (green) maintains a more balanced behavior in postural balance, while it tends to be more erratic in DG (blue) and more noticeably in DNG (orange), which can be attributed to the increase in the deterioration of the systems involved in the balance. This implies that the underlying model allows the generation of appropriate features to be used in the classification models due to the low overlapping data.

In the next inspection, dataset is grouped into two types of evaluations: all tests with eyes open and the ones with eyes closed. The box plots in Fig. 6 shows that, in the set of eyes open, there is a similar distribution between HG and DG on the X axis, unlike DNG that reaches

Table 3

Results of the classification models (Hold-out) by classes.

Model	Accuracy	Error	Precision	Recall	F-Score
Training					
ANN	99.1%	0.9%	99.1%	99.1%	99.1%
K-NN	99%	1%	99%	99%	99%
DTree	97.9%	2.1%	97.9%	97.9%	97.9%
NB	97.7%	2.3%	97.9%	97.9%	97.9%
Testing					
ANN	99.2%	0.8%	99.2%	99.2%	99.2%
K-NN	98.7%	1.3%	98.7%	98.7%	98.7%
DTree	97.1%	2.9%	97.1%	97.1%	97.1%
NB	97.1%	2.9%	97.2%	97.1%	97.1%

Table 4

Results of the testing set in classification models by classes, trained with 10 folds Cross-validation.

Model	Accuracy	Error	Precision	Recall	F-Score
ANN	98.9%	1.1%	98.9%	98.9%	98.9%
K-NN	98.5%	1.5%	98.6%	98.5%	98.5%
DTree	97.7%	2.3%	97.8%	97.7%	97.8%
NB	97.5%	2.5%	97.5%	97.5%	97.5%

higher quartiles (CoP X + EO). However, in derived features ((ACC X, JERK X) + EO), the groups in X axis are separated better than previous one (in a staggered fashion), except in Y axis ((CoP Y, ACC Y, JERK Y) + EO), where just HG is separated. In the case of Fig. 7, in the set of eyes closed, staggered data distribution is obtained in derived features, in both axis, which implies that the distractors highlight the differences in balance between groups. This also shows the progressive deterioration of the diabetic patient's health.

3.2. Classification by classes

This model follows the same approach as Section 2.7, but it considers to classify by classes, including eyes open (EO), eyes closed (EC), unstable surface (US) and dual task (DT) in the same set, that is, for training set the average curve is fitted over all time series, regardless the type of test condition. Regarding the above, each average curve is labeled with its corresponding class value (healthy, diabetic, neuropathic). Results obtained from the execution of algorithms are evaluated using accuracy, error, precision, recall, and F-score metrics (see Table 3). All models achieve an accuracy over 97%. The NB and DTree models achieve 97.3% of accuracy in the testing set. The ANN and K-NN models exceed the previous ones, where ANN achieves an accuracy of 99.2%, close to 98.7% of the K-NN model, in the testing set. This means that the models do not require additional settings. The results of the models are summarized in Table 3, separated in training and testing results.

To prevent the model from being overfitted to the training data, a Cross-Validation method with 10 folds and stratified sampling is used. The results in Table 4 correspond to the testing set. They show that accuracy stays close to 99% in the ANN and K-NN models, while in the NB and DTree models show no significant variations.

The DTree model generates a graphical representation of the features hierarchy, as shown in Fig. 8, where each branch (feature) has an RMS threshold value that separates the classes. The acceleration on X is the one that best discriminates against healthy subjects if RMS acceleration on X is less than or equal to 0.363, followed by the acceleration on Y, which classifies subjects as neuropathic if RMS value is greater than 0.795. Finally, DTree uses COP X to differentiate diabetic subjects from healthy ones. The next leaves of the tree are pruned because they reduce the yield of the tree. All these patterns found and represented by the decision tree model are consistent with the corresponding ones perceived visually from the box plot diagrams shown in Figs. 6 and 7. A decision tree combines, in a hierarchical way,

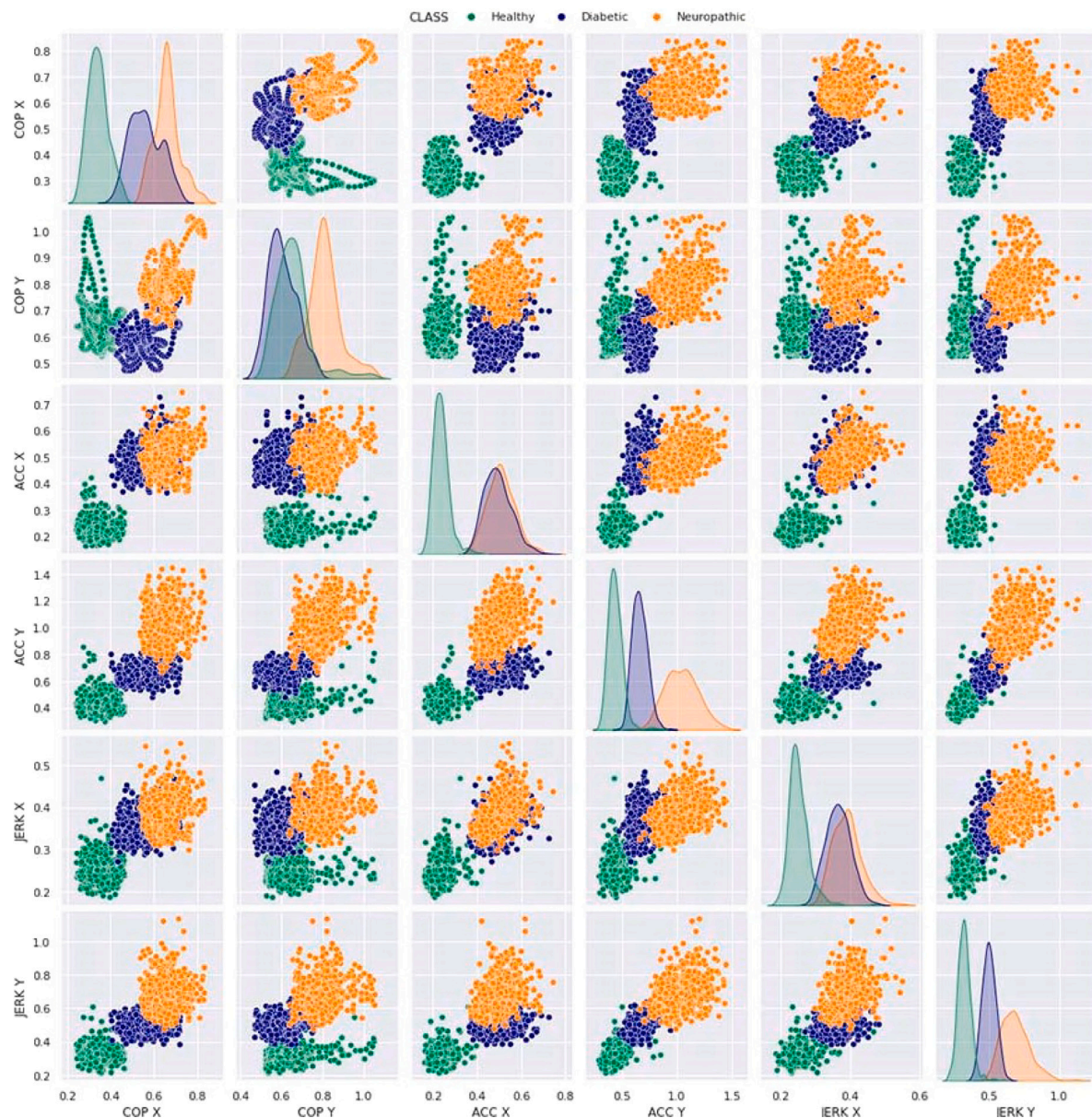


Fig. 5. Pair plot of generated features by the underlying proposed model. Diagonal plots are marginal distributions between classes. Note that ACC Y and JERK Y seem to be good discriminators, based on the diagonal plots. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

features to build a predictive model that has a graphical representation. Data visualizations (e.g., scatterplots, box plots), on the other hand, while providing insight about interesting facts in the data, do not represent an explicit predictive model that can be used directly and automatically to classify new cases.

3.3. Classification by tests

In the classification by test model, the average curve is fitted for each of the eight tests separately. This means, eight data sets are generated (EO, EODT, EOUS, EOUSD, EC, ECDT, ECUS and ECUSD). Each one is divided into 70% for training set and the remaining 30% for testing set. The purpose is to check which set has a positive or negative influence in the performance of the algorithms. That is, which ones make the average curve more complex for classification. Due to the large number of results (32 results, from 8 datasets and 4 machine learning algorithms), Table 5 shows only the ones with the highest performance obtained by each set, using the four machine learning algorithms previously described. ANN and K-NN are the best

with an accuracy of 99% and 95% respectively. The models with the lowest performance were not included in the tables. In the training and testing phase, the EO and EC sets show the best results, over 95.7% accuracy rate. As more distractors are included, classification performance decreases. In testing, ECUS, EOUSD, and EOUS sets show the worst results, under 90% accuracy. This implies that the US (unstable surface) generates more noise in COP than DT (dual task) and EC (eyes closed), thus making the classification task more complex. In other words, complex test conditions, like US, make more difficult the task of discriminating among HG, DG and DNG based on time series analysis for a machine learning algorithm. Though, it does not imply that, for other purposes, conditions like US may be good discriminators.

The same process is repeated, this time training the machine learning algorithms with Cross-Validation to each set of tests, to decrease the probability of overfitting because of random partition of the holdout method. The results in Table 6 show that EO and EC sets perform better and models performance decay as more distractors are included. In this case, the EOUS set shows the worst performance, with an 82.3% of accuracy rate.

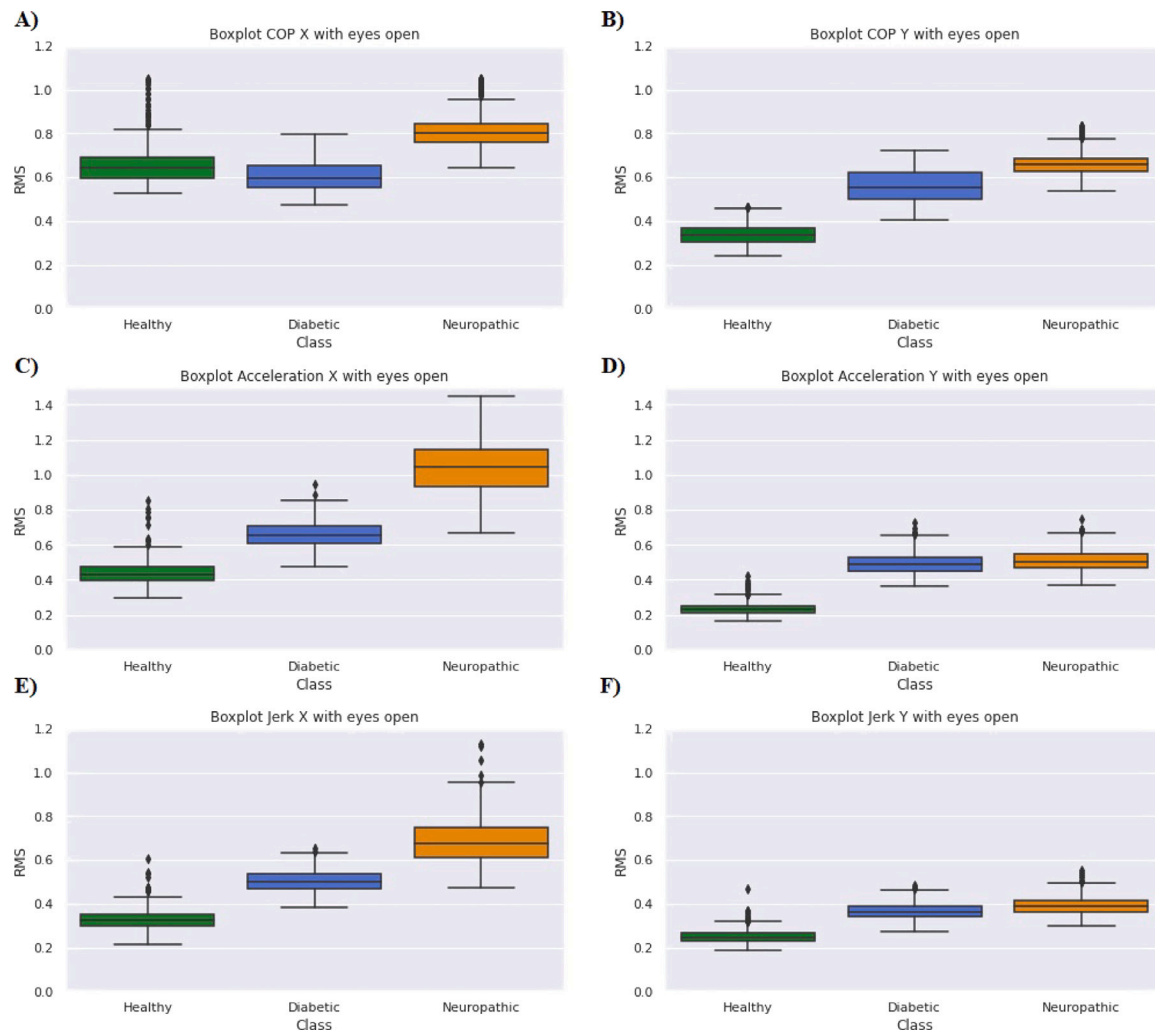


Fig. 6. Box plot of dataset with eyes open (EO) in COP X (A), COP Y (B), Acceleration in X (C), Acceleration in Y (D), Jerk in X (E) and Jerk in Y (F).

Table 5

Results of classification models (Hold-out) by tests.

Model	Exam	Accuracy	Error	Precision	Recall	F-Score
Training						
ANN	EO	99.1%	0.9%	99.1%	99.1%	99.1%
K-NN	EC	97.7%	2.3%	97.7%	97.7%	97.7%
K-NN	ECDT	96.9%	3.1%	96.9%	96.9%	96.9%
K-NN	ECUS	94.7%	5.3%	94.7%	94.7%	94.7%
K-NN	ECUSDT	93.3%	6.7%	93.4%	93.3%	93.3%
K-NN	EODT	93.1%	6.9%	93.1%	93.1%	93.1%
K-NN	EOUSDT	90.7%	9.3%	90.9%	90.7%	90.8%
K-NN	EOUS	87.1%	12.9%	87.3%	87.1%	87.2%
Testing						
ANN	EO	99.2%	0.8%	99.2%	99.2%	99.2%
ANN	EC	95.7%	4.3%	96%	95.7%	95.9%
K-NN	ECDT	94.4%	5.6%	94.4%	94.4%	94.4%
K-NN	ECUSDT	93.3%	6.7%	93.4%	93.3%	93.3%
K-NN	EODT	90.4%	9.6%	90.7%	90.4%	90.6%
K-NN	ECUS	89.9%	10.1%	90%	89.9%	89.9%
K-NN	EOUSDT	86.1%	13.9%	86.8%	86.1%	86.5%
K-NN	EOUS	85.3%	14.7%	85.8%	85.3%	85.5%

In summary, in this section, 32 models were generated, each with its own training and testing set, although only 8 are shown, those with the best performance for each condition. Therefore, the results shown under conditions with distractors (unstable and dual task) were tested with the corresponding test sets for that condition. The same for the

Table 6

Results of classification models by tests, using Cross-validation.

Model	Exam	Accuracy	Error	Precision	Recall	F-Score
ANN	EO	98.7%	1.3%	98.7%	98.7%	98.7%
K-NN	EC	97.1%	2.9%	97.3%	97.1%	97.2%
K-NN	ECDT	95.1%	4.9%	95.3%	95.1%	95.2%
K-NN	ECUS	92.7%	7.3%	92.7%	92.7%	92.7%
K-NN	EODT	91.1%	8.9%	91.3%	91.1%	91.2%
NB	ECUSDT	89.3%	10.7%	89.7%	89.3%	89.5%
ANN	EOUSDT	84.4%	15.6%	85.3%	84.4%	84.8%
ANN	EOUS	84%	16%	84%	84%	84%

case of the EO and EC conditions: their results show the predictive rate for those conditions, they were not tested with other more complex conditions (ECUS, EOUSDT and EOUS). This is because the purpose of this experiment was to see empirically how each condition affects the performance of a classifier, in isolation from the rest of the conditions. In this way, a decrease in the performance of a classifier is very likely to be due to the more complex conditions inherent to the data set and not to an overfitting of a trained model to recognize patterns under simpler conditions (EO, EC).

4. Discussion

The aim of this research was to study pathology detection and medical diagnosis through the analysis of individual postural control

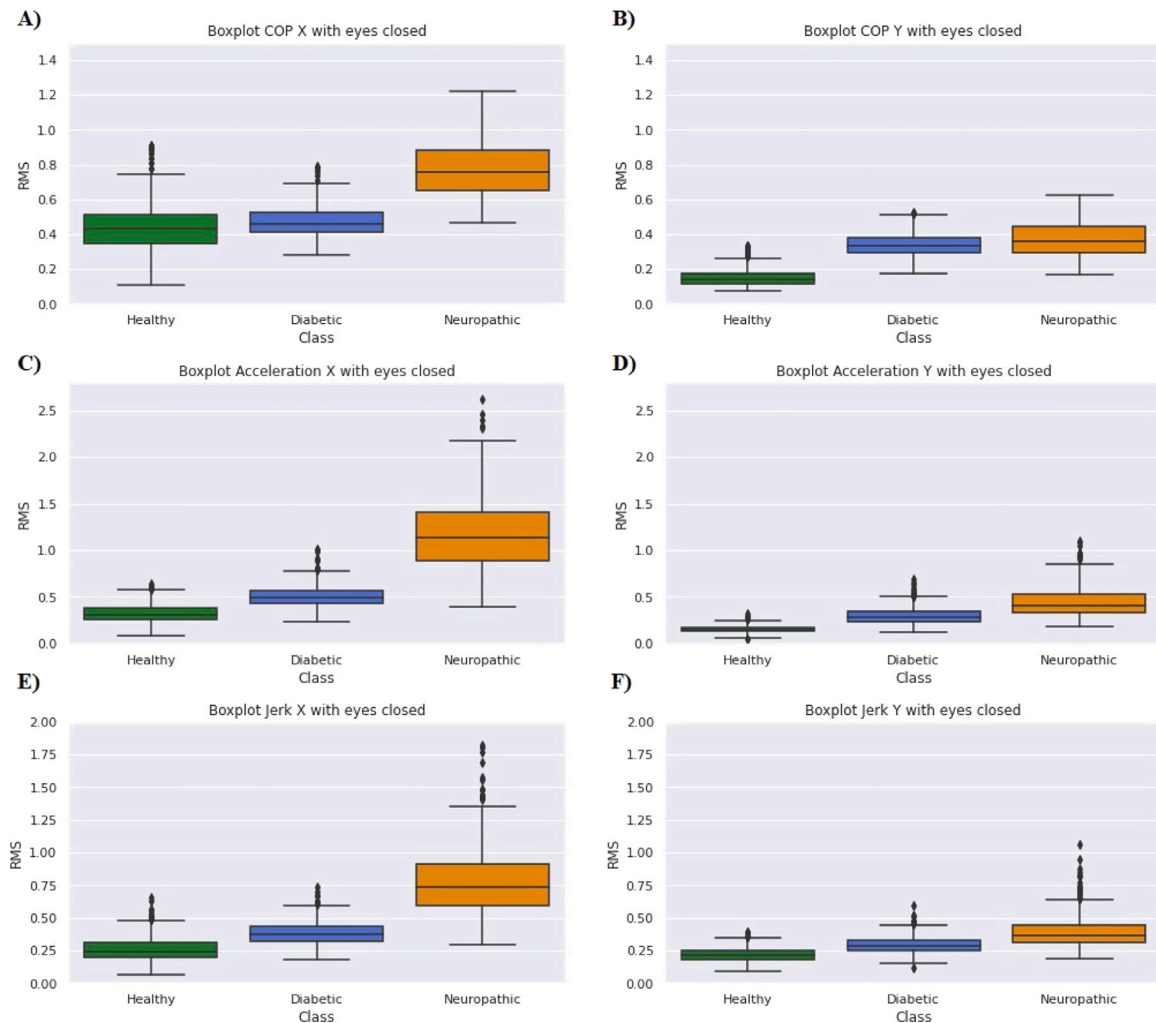


Fig. 7. Box plot of dataset with eyes closed (EC) in COP X (A), COP Y (B), Acceleration in X (C), Acceleration in Y (D), Jerk in X (E) and Jerk in Y (F).

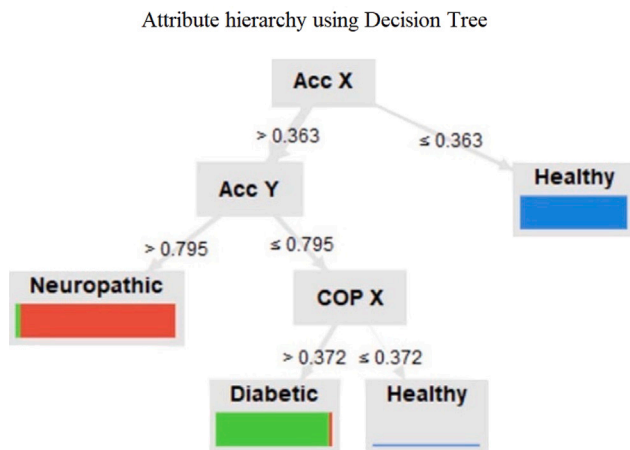


Fig. 8. The decision tree generated for the classification by classes. Acceleration on X (Acc X) is relevant to identify HG, Acceleration Y (Acc Y) for DNG and COP X for DG.

variables, by using machine learning algorithms. The development of a statistical-computational method to classify participants, based on the profile of the signal generated during the postural test, is supported by the analysis of the derived characteristics, since differences are found between participants. An underlying probabilistic model formulated

from the data to feed different machine learning algorithms, facilitates the learning of the classifiers. Their performance is given by the probabilistic model applied to the features over the hyperparameters configured in the algorithms. Nevertheless, the development of separated models by the class and condition of the test suggest that the postural evaluation conditions influence their performance. The generation of empirical evidence, exposed in Section 3, supports the hypothesis that data from the time series allows classifying, with a high level of confidence, the condition of a subject undergoing a postural test.

According to the suggestions of Audiffren [33], SWARII resampling algorithm allows correcting the frequency and removing outliers generated by the WBB. This allows the truncation procedure generates a set of corrected time series of the same length. In postural tests, data truncation around a 20-second window is usually applied when CoP is used to extract parameters. The reason behind this is that, on the one hand, the initial effects of the task are discarded and, on the other hand, including effects related to fatigue are avoided. In addition, this allows to prevent the model from being conditioned to the time series length. Then, new features can be derived in the same time domain for all participants. Therefore, SWARII is considered a useful method to deal with the limitations of the WBB device, particularly when the greatest amount of information about behaviors is in the derivatives. Thus, noise can be reduced without losing more information.

Analysis performed on datasets per postural balance test show that there are test conditions that have a negative influence on postural

balance, which affect classifier performance. The fact that open and closed eyes datasets are always in first place means that unstable surface and dual task make the time series more complex in diabetic subjects, including white noise instead of highlighting unique behaviors in each class. In the case of eyes closed and dual task, accuracy rate exceeds 95%. Therefore, these test conditions manage to increase exclusive behaviors, unlike the unstable surface and the combination of the three conditions together, which tend to generate a more erratic posture. Therefore, perturbations that reducing performance may be discarded to reduce participants' measurement costs.

Through the functional posture data analysis, it is assumed that postural sway is affected more by the neuropathy than by diabetes. This suggest that a person suffering from a pathology at a peripheral nervous system level will show greater signal variability on the Y coordinate axis (i.e., anteroposterior direction), unlike a healthy subject, where distribution remains more stable. This trend has been previously described [41] and was recently taken into consideration in a study conducted by Vargas Matamala et al. (2022), in which, using the WBB, greater magnitudes of COP in the Y-axis were found in subjects with diabetic neuropathy [12]. RMS extraction in the three COP variants (position, acceleration, and jerk) allows separating the groups of healthy, diabetic and neuropathic subjects and estimating similar behaviors for each class. This confirms that, in order to obtain a greater differentiation between classes, feature extraction must be made from derived attributes [7].

Regarding the relevance of the use of machine learning algorithms to classify the participants in one of the three groups, first note that the visually detected patterns constitute interesting facts in the data but in themselves do not constitute a predictive model that can be used direct and automatically to classify new cases, although it is true that they contribute to validating the results of the models, particularly when the model has a graphical representation. Given the exploratory nature of the research, with a more general than specific hypothesis, it was decided to use machine learning algorithms that allow classification models to be directly generated. In particular, in the case of the algorithm that induces a decision tree from the data, it also allows the relative importance of the attributes to be prioritized and builds a model understandable by any end user. It is effective that a simple statistical analysis provides elements to highlight differences between the three groups and can be an element of future work more focused on the selection of relevant features than on the classification of the groups.

Postural deficit screening is an integral component of elderly clinical assessment. The proposed method of classifying subjects, based on characteristics derived from the COP, could be helpful for selecting therapeutic strategies that have proven to be useful in the management of diabetes and diabetic neuropathy (exercise based and lifestyle interventions), and for the evaluation of the impact of such interventions according to the characteristics of the individual.

5. Conclusions

Machine learning models manage to obtain 99% maximum accuracy, depending on the dataset analyzed. This implies that performance is based on the quality of the extracted features, over the hyperparameters configured for each algorithm. Therefore, new posture-related characteristics may be included, to provide to the healthcare professionals more information on a subject's postural sway and health conditions. For future work, features related to the COP like path, sample entropy or ellipse area can be included in the current set to complement the information. Concerning machine learning models, data augmentation is required prior to the design of a deep learning model, which considers low-level features to avoid overlapping data between classes, present in some features like acceleration and jerk.

In addition, the following limitations are identified and discussed. Firstly, our research has a reduced sample size; however, to increase

the number of series, the postural evaluations are repeated, at most, three times for each patient, as long as the subject's health conditions do not prevent otherwise. Secondly, the DN assessment method did not allow more specific insights on nerve damage (i.e., damaged fiber) or other factors involved in balance disorders in this population, such as sarcopenia. Future studies should consider the use of screening tools to assess functional compromise of sensory fibers and loss of skeletal muscle mass to determine their contribution to postural balance performance. Finally, the use of the WBB also involves some technical limitations as it is not a device designed for clinical purposes, however, recommended processing methods were used to mitigate these constraints.

A repository on GitHub is available to verify data at:

<https://github.com/vidcoo/FDA>

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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